

Calculus II

Integrals of involving radicals of quadratics

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Outline

- 1 Integrals of form $\int R(x, \sqrt{ax^2 + bx + c})dx$, R - rational function
 - Transforming to the forms $\sqrt{x^2 + 1}$, $\sqrt{-x^2 + 1}$, $\sqrt{x^2 - 1}$
 - Table of Euler and trig substitutions
 - The case $\sqrt{x^2 + 1}$
 - The case $\sqrt{-x^2 + 1}$
 - The case $\sqrt{x^2 - 1}$

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 - The case $\sqrt{x^2 - 1}$
- 2 Rationalizing Substitutions

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Integrals of form $\int R(x, \sqrt{ax^2 + bx + c})dx$, R - rational function

Let $R(x, y)$ be an arbitrary rational expression in two variables (quotient of polynomials in two variables).

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- We motivate why we need **such integrals** by examples such as computing the area of an ellipse.

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- With $x = a \sin \theta$, the old variable is a function of the new one.

Linear substitutions to simplify radicals $\sqrt{ay^2 + by + c}$

- Using linear substitutions, radicals of form $\sqrt{ay^2 + by + c}$, $a \neq 0$, $b^2 - 4ac \neq 0$ can be transformed to (multiple of):
 - $\sqrt{x^2 + 1}$
 - $\sqrt{-x^2 + 1}$
 - $\sqrt{x^2 - 1}$.
- We already studied how to do that using completing the square when dealing with rational functions.

Recall: linear substitution is subst. of the form $u = px + q$.

Example

Use linear substitution to transform $\sqrt{x^2 + x + 1}$ to multiple of $\sqrt{u^2 + 1}$.

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Recall: linear substitution is subst. of the form $u = px + q$.

Example

Use linear substitution to transform $\sqrt{x^2 + x + 1}$ to multiple of $\sqrt{u^2 + 1}$.

$$\sqrt{x^2 + x + 1} = \sqrt{x^2 + 2 \cdot \frac{1}{2}x + ? - ? + 1}$$

Recall: linear substitution is subst. of the form $u = px + q$.

Example

Use linear substitution to transform $\sqrt{x^2 + x + 1}$ to multiple of $\sqrt{u^2 + 1}$.

$$\sqrt{x^2 + x + 1} = \sqrt{x^2 + 2 \cdot \frac{1}{2}x + \frac{1}{4} - \frac{1}{4} + 1}$$

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Example

Use linear substitution to transform $\sqrt{x^2 + x + 1}$ to multiple of $\sqrt{u^2 + 1}$.

$$\begin{aligned}\sqrt{x^2 + x + 1} &= \sqrt{x^2 + 2 \cdot \frac{1}{2}x + \frac{1}{4} - \frac{1}{4} + 1} \\ &= \sqrt{\left(x + ?\right)^2 + ?}\end{aligned}$$

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Use linear substitution to transform $\sqrt{x^2 + x + 1}$ to multiple of $\sqrt{u^2 + 1}$.

$$\begin{aligned}\sqrt{x^2 + x + 1} &= \sqrt{x^2 + 2 \cdot \frac{1}{2}x + \frac{1}{4} - \frac{1}{4} + 1} \\ &= \sqrt{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}}\end{aligned}$$

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$$\begin{aligned}
 \sqrt{x^2 + x + 1} &= \sqrt{x^2 + 2 \cdot \frac{1}{2}x + \frac{1}{4} - \frac{1}{4} + 1} \\
 &= \sqrt{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} \\
 &= \sqrt{\frac{3}{4} \left(\frac{4}{3} \left(x + \frac{1}{2}\right)^2 + 1\right)}
 \end{aligned}$$

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 &= \sqrt{\frac{3}{4} \left(\frac{4}{3} \left(x + \frac{1}{2}\right)^2 + 1\right)} \\
 &= \frac{\sqrt{3}}{2} \sqrt{\left(\frac{2}{\sqrt{3}} \left(x + \frac{1}{2}\right)\right)^2 + 1}
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 &= \frac{\sqrt{3}}{2} \sqrt{u^2 + 1},
 \end{aligned}$$

$$\text{where } u = \frac{2}{\sqrt{3}} \left(x + \frac{1}{2}\right) = \frac{2\sqrt{3}}{3}x + \frac{\sqrt{3}}{3}.$$

Recall: linear substitution is subst. of the form $u = px + q$.

Example

Use linear subst. to transform $\sqrt{-2x^2 + x + 1}$ to multiple of $\sqrt{-u^2 + 1}$.

$$\sqrt{-2x^2 + x + 1} =$$

Recall: linear substitution is subst. of the form $u = px + q$.

Example

Use linear subst. to transform $\sqrt{-2x^2 + x + 1}$ to multiple of $\sqrt{-u^2 + 1}$.

$$\sqrt{-2x^2 + x + 1} = \sqrt{-2\left(x^2 - \frac{1}{2}x - \frac{1}{2}\right)}$$

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Use linear subst. to transform $\sqrt{-2x^2 + x + 1}$ to multiple of $\sqrt{-u^2 + 1}$.

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 &= \sqrt{\frac{9}{8}\left(-\frac{16}{9}\left(x - \frac{1}{4}\right)^2 + 1\right)}
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- Each of the above integrals can be transformed to a rational trigonometric integral using 3 pairs of substitutions:
 $x = \tan \theta$, $x = \cot \theta$; $x = \sin \theta$, $x = \cos \theta$; $x = \csc \theta$, $x = \sec \theta$.

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 $\int R(x, \sqrt{x^2 + 1})dx$, $\int R(x, \sqrt{-x^2 + 1})dx$, $\int R(x, \sqrt{x^2 - 1})dx$.
- Each of the above integrals can be transformed to a rational trigonometric integral using 3 pairs of substitutions:
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- The Euler substitutions directly transform the integral to a rational function integral.
- We will demonstrate that the Euler substitutions are **rational**.

Trigonometric substitution and Euler substitution

Expression	Substitution	Variable range	Relevant identity
$\sqrt{x^2 + 1}$	$x = \tan \theta$	$\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$1 + \tan^2 \theta = \sec^2 \theta$
	$x = \cot \theta$	$\theta \in (0, \pi)$	$1 + \cot^2 \theta = \csc^2 \theta$
$\sqrt{-x^2 + 1}$	$x = \sin \theta$	$\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$1 - \sin^2 \theta = \cos^2 \theta$
	$x = \cos \theta$	$\theta \in (0, \pi)$	$1 - \cos^2 \theta = \sin^2 \theta$
$\sqrt{x^2 - 1}$	$x = \csc \theta$	$\theta \in \left[0, \frac{\pi}{2}\right) \cup \left[\pi, \frac{3\pi}{2}\right)$	$\csc^2 \theta - 1 = \cot^2 \theta$
	$x = \sec \theta$	$\theta \in \left[0, \frac{\pi}{2}\right) \cup \left[\pi, \frac{3\pi}{2}\right)$	$\sec^2 \theta - 1 = \tan^2 \theta$

Euler substitution by applying in addition $\theta = 2 \arctan t$

$\sqrt{x^2 + 1}$	$x = \frac{2t}{1-t^2}$	$-1 < t < 1$	(?)
	$x = \frac{1}{2} \left(\frac{1}{t} - t \right)$	$0 < t$	(?)
$\sqrt{-x^2 + 1}$	$x = \frac{2t}{1+t^2}$	$-1 \leq t \leq 1$	(?)
	$x = \frac{1-t^2}{1+t^2}$	$0 < t$	(?)
$\sqrt{x^2 - 1}$	$x = \frac{1}{2} \left(\frac{1}{t} + t \right)$	$t \in (-\infty, -1) \cup [0, 1)$	(?)
	$x = \frac{1+t^2}{1-t^2}$	$t \in (-\infty, -1) \cup [0, 1)$	(?)

Trigonometric substitution $x = \cot \theta$ for $\sqrt{x^2 + 1}$

The trigonometric substitution $x = \cot \theta$, $\theta \in (0, \pi)$ for $\sqrt{x^2 + 1}$:

$$\sqrt{x^2 + 1} =$$

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$$\begin{aligned}\sqrt{x^2 + 1} &= \sqrt{\cot^2 \theta + 1} \\ &= \sqrt{\frac{\cos^2 \theta}{\sin^2 \theta} + 1}\end{aligned}$$

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 &= \sqrt{\frac{1}{\sin^2 \theta}} = \frac{1}{\sqrt{\sin^2 \theta}} \\
 &= \frac{1}{\sin \theta}
 \end{aligned}$$

when $\theta \in (0, \pi)$ we have
 $\sin \theta \geq 0$ and so
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The trigonometric substitution $x = \cot \theta$, $\theta \in (0, \pi)$ for $\sqrt{x^2 + 1}$ is given by:

$$\begin{aligned} x &= \cot \theta \\ \sqrt{x^2 + 1} &= \frac{1}{\sin \theta} = \csc \theta \\ dx &= \quad \quad \quad ?d\theta \\ \theta &= \operatorname{arccot} x . \end{aligned}$$

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Example

$$\int \frac{1}{x^2 \sqrt{x^2 + 9}} dx$$

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 \int \frac{1}{x^2 \sqrt{x^2 + 9}} dx &= \int \frac{1}{x^2 3 \sqrt{\left(\frac{x}{3}\right)^2 + 1}} dx \\
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 &= \int \frac{1}{\textcolor{red}{27} \cot^2 \theta \sqrt{\textcolor{red}{?}}} \left(\textcolor{red}{?} \right) d\theta
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 &= \frac{1}{9} \int \frac{-\sin \theta}{\cos^2 \theta} d\theta = \frac{1}{9} \int \frac{1}{\cos^2 \theta} d(?)
 \end{aligned}$$

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 &= \frac{1}{9} \int \frac{-\csc^2 \theta}{\cot^2 \theta \csc \theta} d\theta \\
 &= \frac{1}{9} \int \frac{-\sin \theta}{\cos^2 \theta} d\theta = \frac{1}{9} \int \frac{1}{\cos^2 \theta} d(\cos \theta)
 \end{aligned}$$

Set

$$\frac{x}{3} = \cot \theta$$

$$x = 3 \cot \theta$$

$$\theta \in (0, \pi)$$

$$\theta \in (0, \pi) \Rightarrow \csc \theta > 0$$

Example

$$\begin{aligned}
 \int \frac{1}{x^2 \sqrt{x^2 + 9}} dx &= \int \frac{1}{x^2 3 \sqrt{\left(\frac{x}{3}\right)^2 + 1}} dx \\
 &= \int \frac{1}{(3 \cot \theta)^2 3 \sqrt{\cot^2 \theta + 1}} d(3 \cot \theta) \\
 &= \int \frac{1}{27 \cot^2 \theta \sqrt{\csc^2 \theta}} (-3 \csc^2 \theta) d\theta \\
 &= \frac{1}{9} \int \frac{-\csc^2 \theta}{\cot^2 \theta \csc \theta} d\theta \\
 &= \frac{1}{9} \int \frac{-\sin \theta}{\cos^2 \theta} d\theta = \frac{1}{9} \int \frac{1}{\cos^2 \theta} d(\cos \theta) \\
 &= \frac{1}{9} \int \frac{du}{u^2}
 \end{aligned}$$

Set

$\frac{x}{3} = \cot \theta$
 $x = 3 \cot \theta$
 $\theta \in (0, \pi)$
 $\theta \in (0, \pi) \Rightarrow \csc \theta > 0$

Set $u = \cos \theta$

Example

$$\begin{aligned}
 \int \frac{1}{x^2 \sqrt{x^2 + 9}} dx &= \int \frac{1}{x^2 3 \sqrt{\left(\frac{x}{3}\right)^2 + 1}} dx \\
 &= \int \frac{1}{(3 \cot \theta)^2 3 \sqrt{\cot^2 \theta + 1}} d(3 \cot \theta) \\
 &= \int \frac{1}{27 \cot^2 \theta \sqrt{\csc^2 \theta}} (-3 \csc^2 \theta) d\theta \\
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 &= \frac{1}{9} \int \frac{-\sin \theta}{\cos^2 \theta} d\theta = \frac{1}{9} \int \frac{1}{\cos^2 \theta} d(\cos \theta) \\
 &= \frac{1}{9} \int \frac{du}{u^2} = ? + C
 \end{aligned}$$

Set

$$\frac{x}{3} = \cot \theta$$

$$x = 3 \cot \theta$$

$$\theta \in (0, \pi)$$

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$$\text{Set } u = \cos \theta$$

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 &= \int \frac{1}{27 \cot^2 \theta \sqrt{\csc^2 \theta}} (-3 \csc^2 \theta) d\theta \\
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 &= \frac{1}{9} \int \frac{du}{u^2} = -\frac{1}{9u} + C
 \end{aligned}$$

Set

$$\frac{x}{3} = \cot \theta$$

$$x = 3 \cot \theta$$

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 \end{aligned}$$

Set

$$\frac{x}{3} = \cot \theta$$

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$$\theta \in (0, \pi)$$

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Example

$$\begin{aligned}
 \int \frac{1}{x^2 \sqrt{x^2 + 9}} dx &= \int \frac{1}{x^2 3 \sqrt{\left(\frac{x}{3}\right)^2 + 1}} dx \\
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Example

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 \end{aligned}$$

Set

$$\frac{x}{3} = \cot \theta$$

$$x = 3 \cot \theta$$

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Example

$$\begin{aligned}
 \int \frac{1}{x^2 \sqrt{x^2 + 9}} dx &= \int \frac{1}{x^2 3 \sqrt{\left(\frac{x}{3}\right)^2 + 1}} dx \\
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 &= \int \frac{1}{27 \cot^2 \theta \sqrt{\csc^2 \theta}} (-3 \csc^2 \theta) d\theta \\
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 &= \frac{1}{9} \int \frac{-\sin \theta}{\cos^2 \theta} d\theta = \frac{1}{9} \int \frac{1}{\cos^2 \theta} d(\cos \theta) \\
 &= \frac{1}{9} \int \frac{du}{u^2} = -\frac{1}{9u} + C = -\frac{\sec \theta}{9} + C \\
 &= -\frac{\sqrt{x^2 + 9}}{9x} + C
 \end{aligned}$$

Set

$$\frac{x}{3} = \cot \theta$$

$$x = 3 \cot \theta$$

$$\theta \in (0, \pi)$$

$$\theta \in (0, \pi) \Rightarrow \csc \theta > 0$$

Set $u = \cos \theta$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t$, transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t$, transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above?

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? **We get the Euler substitution:**

$$x =$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \cot \theta$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$\begin{aligned}x &= \cot \theta \\&= \cot (2 \arctan t)\end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \cot \theta$$

$$= \cot(2 \arctan t) \quad \left| \text{Recall: } \cot(2z) = \frac{\cos(2z)}{\sin(2z)} \right.$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \cot \theta$$

$$= \cot(2 \arctan t) \quad \left| \text{Recall: } \cot(2z) = \frac{\cos(2z)}{\sin(2z)} = \frac{1 - \tan^2 z}{2 \tan z} \right.$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \cot \theta$$

$$\begin{aligned}
 &= \cot(2 \arctan t) & | \text{Recall: } \cot(2z) &= \frac{\cos(2z)}{\sin(2z)} = \frac{1 - \tan^2 z}{2 \tan z} \\
 &= \frac{1 - \tan^2(\arctan t)}{2 \tan(\arctan t)}
 \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
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 &= \frac{1 - \tan^2(\arctan t)}{2 \tan(\arctan t)} \\
 &= \frac{1 - t^2}{2t}
 \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
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 &= \frac{1 - \tan^2(\arctan t)}{2 \tan(\arctan t)} \\
 &= \frac{1 - t^2}{2t} \\
 &= \frac{1}{2} \left(\frac{1}{t} - t \right) .
 \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

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 &= \frac{1 - \tan^2(\arctan t)}{2 \tan(\arctan t)} \\
 &= \frac{1 - t^2}{2t} \\
 &= \frac{1}{2} \left(\frac{1}{t} - t \right) .
 \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
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 &= \frac{1 - t^2}{2t} \\
 &= \frac{1}{2} \left(\frac{1}{t} - t \right) .
 \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
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 &= \frac{1 - \tan^2(\arctan t)}{2 \tan(\arctan t)} \\
 &= \frac{1 - t^2}{2t} \\
 &= \frac{1}{2} \left(\frac{1}{t} - t \right) .
 \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\sqrt{x^2 + 1} =$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\sqrt{x^2 + 1} = \sqrt{\frac{1}{4} \left(\frac{1}{t} - t \right)^2 + 1}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\begin{aligned} \sqrt{x^2 + 1} &= \sqrt{\frac{1}{4} \left(\frac{1}{t} - t \right)^2 + 1} \\ &= \frac{1}{2} \sqrt{\left(\frac{1}{t} - t \right)^2 + 4} \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\begin{aligned} \sqrt{x^2 + 1} &= \sqrt{\frac{1}{4} \left(\frac{1}{t} - t \right)^2 + 1} \\ &= \frac{1}{2} \sqrt{\left(\frac{1}{t} - t \right)^2 + 4} \quad \mid \quad \left(\frac{1}{t} - t \right)^2 + 4 = \left(\frac{1}{t} + t \right)^2 \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\begin{aligned} \sqrt{x^2 + 1} &= \sqrt{\frac{1}{4} \left(\frac{1}{t} - t \right)^2 + 1} \\ &= \frac{1}{2} \sqrt{\left(\frac{1}{t} - t \right)^2 + 4} \quad \Bigg| \quad \left(\frac{1}{t} - t \right)^2 + 4 = \left(\frac{1}{t} + t \right)^2 \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
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- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
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$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\begin{aligned} \sqrt{x^2 + 1} &= \sqrt{\frac{1}{4} \left(\frac{1}{t} - t \right)^2 + 1} \\ &= \frac{1}{2} \sqrt{\left(\frac{1}{t} + t \right)^2} \quad \left| \begin{array}{l} \sqrt{\left(\frac{1}{t} + t \right)^2} = \frac{1}{t} + t \\ \text{because } t > 0 \end{array} \right. \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
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$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\begin{aligned} \sqrt{x^2 + 1} &= \sqrt{\frac{1}{4} \left(\frac{1}{t} - t \right)^2 + 1} \\ &= \frac{1}{2} \sqrt{\left(\frac{1}{t} + t \right)^2} \quad \left| \begin{array}{l} \sqrt{\left(\frac{1}{t} + t \right)^2} = \frac{1}{t} + t \\ \text{because } t > 0 \end{array} \right. \\ &= \frac{1}{2} \left(\frac{1}{t} + t \right) . \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\begin{aligned} \sqrt{x^2 + 1} &= \sqrt{\frac{1}{4} \left(\frac{1}{t} - t \right)^2 + 1} \\ &= \frac{1}{2} \sqrt{\left(\frac{1}{t} + t \right)^2} \quad \left| \begin{array}{l} \sqrt{\left(\frac{1}{t} + t \right)^2} = \frac{1}{t} + t \\ \text{because } t > 0 \end{array} \right. \\ &= \frac{1}{2} \left(\frac{1}{t} + t \right) . \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\sqrt{x^2 + 1} = \frac{1}{2} \left(\frac{1}{t} + t \right) .$$

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What if we compose the above? We get the Euler substitution:

Definition

The Euler substitution for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$ is given by:

$$\begin{aligned} x &= \frac{1}{2} \left(\frac{1}{t} - t \right), & t > 0 \\ \sqrt{x^2 + 1} &= \frac{1}{2} \left(\frac{1}{t} + t \right) \\ dx &= -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\ t &= \sqrt{x^2 + 1} - x \end{aligned}$$

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Example

$$\int \sqrt{x^2 + 1} \, dx =$$

Euler substitution: $x = \frac{1}{2} \left(\frac{1}{t} - t \right)$, $\sqrt{x^2 + 1} = \frac{1}{2} \left(\frac{1}{t} + t \right)$,
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Example

$$\int \sqrt{x^2 + 1} \, dx = - \int \frac{1}{2} \left(\frac{1}{t} + t \right) \frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt$$

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 &= \frac{1}{2} \left(\frac{1}{2} (t^{-1} - t) \frac{1}{2} (t^{-1} + t) \right) - \frac{1}{2} \ln t + C
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 &= \frac{1}{2} \left(\frac{1}{2} \left(t^{-1} - t \right) \frac{1}{2} \left(t^{-1} + t \right) \right) - \frac{1}{2} \ln t + C \\
 &= \frac{1}{2} x \sqrt{x^2 + 1} - \frac{1}{2} \ln \left(\sqrt{x^2 + 1} - x \right) + C
 \end{aligned}$$

Euler substitution: $x = \frac{1}{2} \left(\frac{1}{t} - t \right)$, $\sqrt{x^2 + 1} = \frac{1}{2} \left(\frac{1}{t} + t \right)$,
 $t = \sqrt{x^2 + 1} - x$, $dx = -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt$. Recall $t > 0$.

Example

$$\begin{aligned}
 \int \sqrt{x^2 + 1} \, dx &= - \int \frac{1}{2} \left(\frac{1}{t} + t \right) \frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\
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Example

Find the area locked b-n the hyperbolas $y = \pm\sqrt{x^2 + 1}$ and $x = \pm 2\sqrt{2}$.

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 $\Rightarrow y^2 - x^2 = 1$ is the **hyperbola**
 $v = \frac{1/2}{u}$ in the (u, v) -plane.

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The area in question is:

$$\int_{?}^{?} 2\sqrt{x^2 + 1} dx$$

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$$\int_{-2\sqrt{2}}^{2\sqrt{2}} 2\sqrt{x^2 + 1} dx$$

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The trigonometric substitution $x = \cos \theta$, $\theta \in [0, \pi]$ for $\sqrt{-x^2 + 1}$:

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 &= \sqrt{\sin^2 \theta} \\
 &= \sin \theta \quad .
 \end{aligned}
 \quad \left| \begin{array}{l} \text{when } \theta \in [0, \pi] \text{ we have} \\ \sin \theta \geq 0 \text{ and so } \sqrt{\sin^2 \theta} = \sin \theta \end{array} \right.$$

To summarize:

Definition

The trigonometric substitution $x = \cos \theta$, $\theta \in [0, \pi]$ for $\sqrt{-x^2 + 1}$ is given by:

$$\begin{aligned}
 x &= \cos \theta \\
 \sqrt{-x^2 + 1} &= \sin \theta \\
 dx &= -\sin \theta d\theta \\
 \theta &= \arccos x \quad .
 \end{aligned}$$

Example

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Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a, b > 0$.

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- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

$$x = \cos \theta$$

$$= \cos(2 \arctan t)$$

$$\left| \cos(2z) = \frac{1 - \tan^2 z}{1 + \tan^2 z} \right.$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$\begin{aligned}
 x &= \cos \theta \\
 &= \cos(2 \arctan t) \\
 &= \frac{1 - \tan^2(\arctan t)}{1 + \tan^2(\arctan t)}
 \end{aligned}
 \quad \left| \quad \cos(2z) = \frac{1 - \tan^2 z}{1 + \tan^2 z}
 \right.$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

$$\begin{aligned}
 x &= \cos \theta \\
 &= \cos(2 \arctan t) & \left| \cos(2z) = \frac{1 - \tan^2 z}{1 + \tan^2 z} \right. \\
 &= \frac{1 - \tan^2(\arctan t)}{1 + \tan^2(\arctan t)} \\
 &= \frac{1 - t^2}{1 + t^2}
 \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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 x &= \cos \theta \\
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What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\sqrt{-x^2 + 1} =$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\sqrt{-x^2 + 1} = \sqrt{1 - \left(\frac{1 - t^2}{1 + t^2} \right)^2}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\begin{aligned} \sqrt{-x^2 + 1} &= \sqrt{1 - \left(\frac{1 - t^2}{1 + t^2}\right)^2} \\ &= \sqrt{\frac{(1 + t^2)^2 - (1 - t^2)^2}{(1 + t^2)^2}} \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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$$\begin{aligned} \sqrt{-x^2 + 1} &= \sqrt{1 - \left(\frac{1 - t^2}{1 + t^2}\right)^2} \\ &= \sqrt{\frac{(1 + t^2)^2 - (1 - t^2)^2}{(1 + t^2)^2}} \quad | \quad (1 + t^2)^2 - (1 - t^2)^2 = ? \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\begin{aligned} \sqrt{-x^2 + 1} &= \sqrt{1 - \left(\frac{1 - t^2}{1 + t^2}\right)^2} \\ &= \sqrt{\frac{(1 + t^2)^2 - (1 - t^2)^2}{(1 + t^2)^2}} \quad | \quad (1 + t^2)^2 - (1 - t^2)^2 = 4t^2 \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\begin{aligned} \sqrt{-x^2 + 1} &= \sqrt{1 - \left(\frac{1 - t^2}{1 + t^2}\right)^2} \\ &= \sqrt{\frac{(1 + t^2)^2 - (1 - t^2)^2}{(1 + t^2)^2}} \quad | \quad (1 + t^2)^2 - (1 - t^2)^2 = 4t^2 \\ &= \sqrt{\frac{4t^2}{(1 + t^2)^2}} \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\begin{aligned} \sqrt{-x^2 + 1} &= \sqrt{1 - \left(\frac{1 - t^2}{1 + t^2}\right)^2} \\ &= \sqrt{\frac{(1 + t^2)^2 - (1 - t^2)^2}{(1 + t^2)^2}} & | (1 + t^2)^2 - (1 - t^2)^2 = 4t^2 \\ &= \sqrt{\frac{4t^2}{(1 + t^2)^2}} & | \sqrt{4t^2} = 2t \text{ because } t > 0 \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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$$(1 + t^2)x = 1 - t^2$$

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- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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$$t^2(x + 1) = 1 - x$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

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$$x = \frac{1 - t^2}{1 + t^2}$$

$$\sqrt{-x^2 + 1} = \frac{2t}{1 + t^2}$$

$$(1 + t^2)x = 1 - t^2$$

$$t^2(\textcolor{red}{x} + \textcolor{red}{1}) = 1 - x$$

$$t^2 = \frac{1 - x}{\textcolor{red}{1} + \textcolor{red}{x}}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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$$t^2(x + 1) = 1 - x$$

$$t^2 = \frac{1 - x}{1 + x}$$

$$t = \frac{\sqrt{1 - x}}{\sqrt{1 + x}}$$

we use $t > 0$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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$$(1 + t^2)x = 1 - t^2$$

$$t^2(x + 1) = 1 - x$$

$$t^2 = \frac{1 - x}{1 + x}$$

$$t = \frac{\sqrt{1 - x} \sqrt{1 + x}}{\sqrt{1 + x} \sqrt{1 + x}} = \frac{\sqrt{-x^2 + 1}}{x + 1}$$

we use $t > 0$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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$$\sqrt{-x^2 + 1} = \frac{2t}{1 + t^2}$$

$$(1 + t^2)x = 1 - t^2$$

$$t^2(x + 1) = 1 - x$$

$$t^2 = \frac{1 - x}{1 + x}$$

$$t = \frac{\sqrt{1 - x}}{\sqrt{1 + x}} = \frac{\sqrt{-x^2 + 1}}{x + 1}$$

we use $t > 0$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\sqrt{-x^2 + 1} = \frac{2t}{1 + t^2}$$

$$t = \frac{\sqrt{-x^2 + 1}}{x + 1}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\sqrt{-x^2 + 1} = \frac{2t}{1 + t^2}$$

$$t = \frac{\sqrt{-x^2 + 1}}{x + 1}$$

dx

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\sqrt{-x^2 + 1} = \frac{2t}{1 + t^2}$$

$$t = \frac{\sqrt{-x^2 + 1}}{x + 1}$$

$$dx = d\left(\frac{1 - t^2}{1 + t^2}\right)$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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$$x = \frac{1 - t^2}{1 + t^2}$$

$$\sqrt{-x^2 + 1} = \frac{2t}{1 + t^2}$$

$$t = \frac{\sqrt{-x^2 + 1}}{x + 1}$$

$$dx = d\left(\frac{1 - t^2}{1 + t^2}\right) = d\left(\frac{2 - (1 + t^2)}{1 + t^2}\right)$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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What if we compose the above? We get the Euler substitution:

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$$\sqrt{-x^2 + 1} = \frac{2t}{1 + t^2}$$

$$t = \frac{\sqrt{-x^2 + 1}}{x + 1}$$

$$\begin{aligned} dx &= d\left(\frac{1 - t^2}{1 + t^2}\right) = d\left(\frac{2 - (1 + t^2)}{1 + t^2}\right) \\ &= d\left(\frac{2}{1 + t^2} - 1\right) \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

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$$x = \frac{1 - t^2}{1 + t^2}$$

$$\sqrt{-x^2 + 1} = \frac{2t}{1 + t^2}$$

$$t = \frac{\sqrt{-x^2 + 1}}{x + 1}$$

$$\begin{aligned} dx &= d\left(\frac{1 - t^2}{1 + t^2}\right) = d\left(\frac{2 - (1 + t^2)}{1 + t^2}\right) \\ &= d\left(\frac{2}{1 + t^2} - 1\right) = -\frac{4t}{(1 + t^2)^2} dt \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
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$$\begin{aligned} dx &= d\left(\frac{1 - t^2}{1 + t^2}\right) = d\left(\frac{2 - (1 + t^2)}{1 + t^2}\right) \\ &= d\left(\frac{2}{1 + t^2} - 1\right) = -\frac{4t}{(1 + t^2)^2} dt \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

Definition

The Euler substitution for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$ is given by:

$$\begin{aligned} x &= \frac{1 - t^2}{1 + t^2}, & t > 0 \\ \sqrt{-x^2 + 1} &= \frac{2t}{1 + t^2} \\ dx &= -\frac{4t}{(t^2 + 1)^2} dt \\ t &= \frac{\sqrt{-x^2 + 1}}{x + 1}. \end{aligned}$$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

$$\sqrt{x^2 - 1} =$$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

$$\sqrt{x^2 - 1} = \sqrt{\sec^2 \theta - 1}$$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

$$\begin{aligned}\sqrt{x^2 - 1} &= \sqrt{\sec^2 \theta - 1} \\ &= \sqrt{\frac{1}{\cos^2 \theta} - 1}\end{aligned}$$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

$$\begin{aligned}\sqrt{x^2 - 1} &= \sqrt{\sec^2 \theta - 1} \\ &= \sqrt{\frac{1}{\cos^2 \theta} - 1} \\ &= \sqrt{\frac{\sin^2 \theta}{\cos^2 \theta}}\end{aligned}$$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

$$\begin{aligned}\sqrt{x^2 - 1} &= \sqrt{\sec^2 \theta - 1} \\ &= \sqrt{\frac{1}{\cos^2 \theta} - 1} \\ &= \sqrt{\frac{\sin^2 \theta}{\cos^2 \theta}} \\ &= \sqrt{\tan^2 \theta}\end{aligned}$$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

$$\begin{aligned}\sqrt{x^2 - 1} &= \sqrt{\sec^2 \theta - 1} \\ &= \sqrt{\frac{1}{\cos^2 \theta} - 1} \\ &= \sqrt{\frac{\sin^2 \theta}{\cos^2 \theta}} \\ &= \sqrt{\tan^2 \theta}\end{aligned}$$

when $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$ we have
 $\tan \theta \geq 0$ and so $\sqrt{\tan^2 \theta} = \tan \theta$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

$$\begin{aligned}
 \sqrt{x^2 - 1} &= \sqrt{\sec^2 \theta - 1} \\
 &= \sqrt{\frac{1}{\cos^2 \theta} - 1} \\
 &= \sqrt{\frac{\sin^2 \theta}{\cos^2 \theta}} \\
 &= \sqrt{\tan^2 \theta} \\
 &= \tan \theta .
 \end{aligned}$$

when $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$ we have
 $\tan \theta \geq 0$ and so $\sqrt{\tan^2 \theta} = \tan \theta$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

$$\begin{aligned}
 \sqrt{x^2 - 1} &= \sqrt{\sec^2 \theta - 1} \\
 &= \sqrt{\frac{1}{\cos^2 \theta} - 1} \\
 &= \sqrt{\frac{\sin^2 \theta}{\cos^2 \theta}} \\
 &= \sqrt{\tan^2 \theta} \\
 &= \tan \theta .
 \end{aligned}$$

when $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$ we have
 $\tan \theta \geq 0$ and so $\sqrt{\tan^2 \theta} = \tan \theta$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

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Find $\int \frac{dx}{\sqrt{x^2 - a^2}}, a > 0$.

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Euler substitution $x = \sec \theta$, $\theta = 2 \arctan t$

- $x = \sec \theta$ transforms $dx, x, \sqrt{x^2 - 1}$ to trig form.
- $\theta = 2 \arctan t$, rationalizes $d\theta, \cos \theta, \sin \theta$.

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What if we compose the above? **We get the Euler substitution:**

$$x =$$

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$$x = -1 + \frac{2}{1 - t^2}$$

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$$\begin{aligned} dx &= d\left(-1 + \frac{2}{1 - t^2}\right) \\ &= ? \end{aligned}$$

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Definition

The Euler substitution for $\sqrt{x^2 - 1}$ corresponding to $x = \sec \theta$ is given by:

$$\begin{aligned} x &= \frac{1 + t^2}{1 - t^2}, & t &\in (-\infty, -1) \cup [0, 1) \\ \sqrt{x^2 - 1} &= \frac{2t}{1 - t^2} \\ dx &= \frac{4t}{(1 - t^2)^2} dt \\ t &= \pm \frac{\sqrt{x^2 - 1}}{x + 1} \end{aligned}$$

Rationalizing Substitutions

Some non-rational fractions can be changed into rational fractions by means of appropriate substitutions. In particular, when an integrand contains an expression of the form $\sqrt[n]{g(x)}$, the substitution $u = \sqrt[n]{g(x)}$ may be effective.

Example

$$\int \frac{\sqrt{x+4}}{x} dx$$

Example

Let $u = \sqrt{x+4}$. Then $u^2 = x+4$, so $x = ?$ and $dx = ?$.

$$\int \frac{\sqrt{x+4}}{x} dx = \int \frac{?}{?} ?$$

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Let $u = \sqrt{x+4}$. Then $u^2 = x+4$, so $x = u^2 - 4$ and $dx = ?$.

$$\int \frac{\sqrt{x+4}}{x} dx = \int \frac{u}{u^2 - 4} ?$$

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Let $u = \sqrt{x+4}$. Then $u^2 = x+4$, so $x = u^2 - 4$ and $dx = 2u du$.

$$\int \frac{\sqrt{x+4}}{x} dx = \int \frac{u}{u^2 - 4} 2u du$$

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Let $u = \sqrt{x+4}$. Then $u^2 = x+4$, so $x = u^2 - 4$ and $dx = 2u du$.

$$\begin{aligned}\int \frac{\sqrt{x+4}}{x} dx &= \int \frac{u}{u^2 - 4} 2u du \\ &= 2 \int \frac{u^2}{u^2 - 4} du\end{aligned}$$

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Let $u = \sqrt{x+4}$. Then $u^2 = x+4$, so $x = u^2 - 4$ and $dx = 2u du$.

$$\begin{aligned}\int \frac{\sqrt{x+4}}{x} dx &= \int \frac{u}{u^2-4} 2u du \\ &= 2 \int \frac{u^2}{u^2-4} du \\ &= 2 \int \left(1 + \frac{4}{u^2-4} \right) du\end{aligned}$$

| long division

Example

Let $u = \sqrt{x+4}$. Then $u^2 = x+4$, so $x = u^2 - 4$ and $dx = 2u du$.

$$\begin{aligned}\int \frac{\sqrt{x+4}}{x} dx &= \int \frac{u}{u^2-4} 2u du \\&= 2 \int \frac{u^2}{u^2-4} du \\&= 2 \int \left(1 + \frac{4}{u^2-4} \right) du && \left| \begin{array}{l} \text{long division} \end{array} \right. \\&= 2 \int du + 8 \int \frac{du}{u^2-4}\end{aligned}$$

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$$\begin{aligned}
 \int \frac{\sqrt{x+4}}{x} dx &= \int \frac{u}{u^2-4} 2u du \\
 &= 2 \int \frac{u^2}{u^2-4} du \\
 &= 2 \int \left(1 + \frac{4}{u^2-4} \right) du && \left| \text{long division} \right. \\
 &= 2 \int du + 8 \int \frac{du}{u^2-4} \\
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 &= 2u + 2(\ln|u-2| - \ln|u+2|) + C \\
 &= 2\sqrt{x+4} + 2 \ln \left| \frac{\sqrt{x+4}-2}{\sqrt{x+4}+2} \right| + C
 \end{aligned}$$